AI-DRIVEN POLYMER AGING PREDICTION USING RAMAN SPECTROSCOPY

PREDICTING POLYMER AGING AND RECYCLABILITY THROUGH MACHINE LEARNING

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AI-DRIVEN POLYMER AGING PREDICTION AND CLASSIFICATION SYSTEM

- Goal: To predict polymer aging from Raman spectra
  - Why it matters (project rationale):
    - Better recycling decisions are possible.
    - Faster material characterization in monomer-chain transfer monomer reaction analysis.
    - Reproducible AI workflow: For IA HPC-bound datasets (to OSC, Case, or Microsoft Azure), using PyTorch, TensorFlow, and conda code can balance the workload of empirically observed reaction values.
    - for input parameters to be computed remotely and not crash/lock up your laptop for 500 input values→5.95 hours on a laptop-based Intel Core Ultra 7 CPU.
    - This code is extensible to NVIDIA GPU CUDA core computation to save computing time of the average medical researcher running PET-RAFT CTM kinetic reaction data analysis: the real magic of artificial intelligence/machine learning based computer programming source code and remote HPC GPU computing in tandem—if and when necessary and appropriate.
  - Long-term Vision:
    - Raman
    - FTIR
    - Image-based models

MODEL ARCHITECTURES

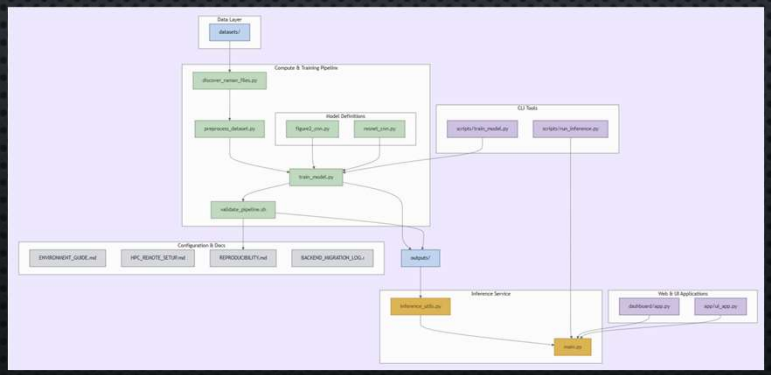
Figure2CNN	ResNet1D	ResNet18Vision
Baseline model from literature.	Deeper candidate spectral model with skip connections.	Future Image-focused CNN architecture model, retrained on polymer dataset (roadmap).

- Future expansions will add additional trained CNNs, supporting direct benchmarking and comparative reporting.

PROJECT STRUCTURE (CLEANED AND CURRENT)

Dataset→ Preprocessing→ Training→ Validation→ Prediction (Output Data To Form a Conclusion)

```
ml-polymer-recycling/
├── datasets/
├── models/              # Model architectures
├── scripts/             # Training, inference, utilities
├── outputs/             # Artifacts: models, logs, plots
├── docs/                # Documentation & reports
└── environment.yml      # (local) Conda execution environment
```



CURRENT STATUS

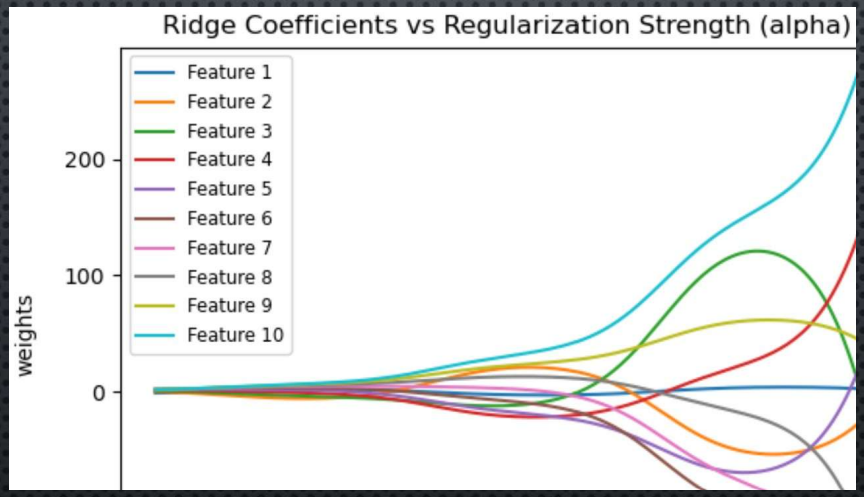
Raman	Active & validated	**87.81% ± 7.59%**
Image	Planned Expansion	N/A
FTIR	Reactivating Modules Soon	N/A

Key Features

- 10-Fold Stratified Cross-Validation
- CLI Training: train_model.py
- CLI Inference run_inference.py
- Output artifact naming per model
- Raman-only preprocessing with baseline correction, smoothing, normalization
- Structured diagnostics JSON (accuracies, confusion matrices)
- Canonical validation script (validate_pipeline.sh) confirms reproducibility of all core components

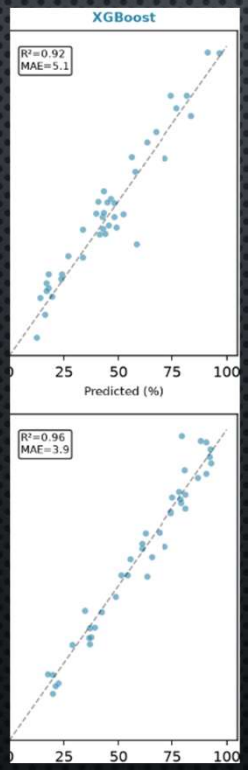
10-FOLD STRATIFIED CROSS-VALIDATION: A FEATURE RIDGE-COEFFICIENT TRACE GRAPH

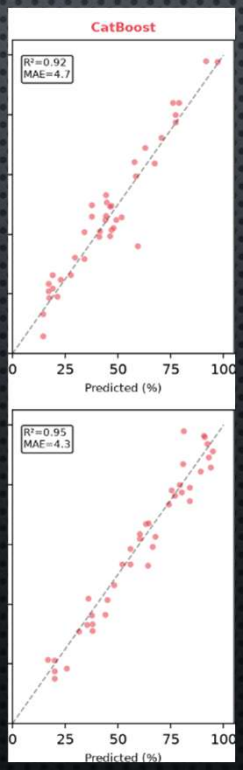
<SCIKIT LINEAR REGRESSION STATISTICS LIBRARY>



- A ridge trace from linear regression analysis reveals that, for small alpha trend analysis, other curve fitting is possible.
- A ridge trace from linear regression analysis reveals that, for large alpha trend analysis, linear chi-squared fitting is most likely.

XGBOOST AND CATBOOST LINEAR REGRESSION COMPARISON





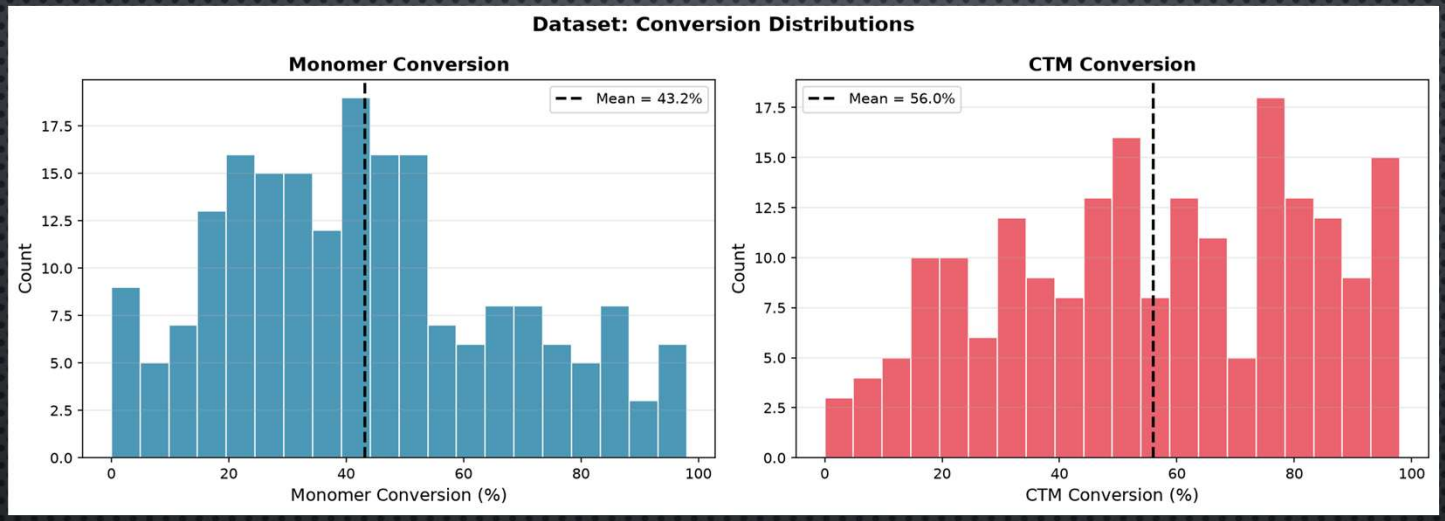
WHAT XGBOOST DOES:

- Handles Tabular Data Best
- Manages missing values
- Prevents curve overfitting
- Captures Non-Linear Relationships
- Features Parallel computing

WHAT CATBOOST DOES:

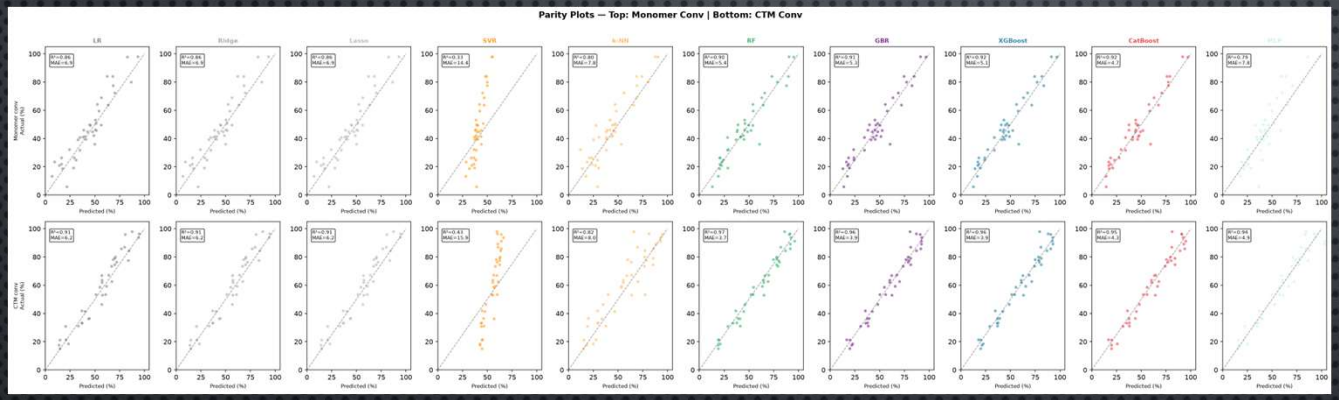
- Native Categorical encoding
- Prevents Target Leakage
- Exceptional default settings
- Ultra-fast real-time inference
- Highly Efficient GPU Training

DATASET: CONVERSION DISTRIBUTIONS



- This is a chart that documents the correlation between reactant monomers and chain transfer monomers into …
- Their observed product yield through their absence after certain timesplice intervals are measured according to …
- The elapsed time transpired afterward (1+, 2+, 3+, 4+, 5+, 6+, and 8+ hours after initial chemical reaction.

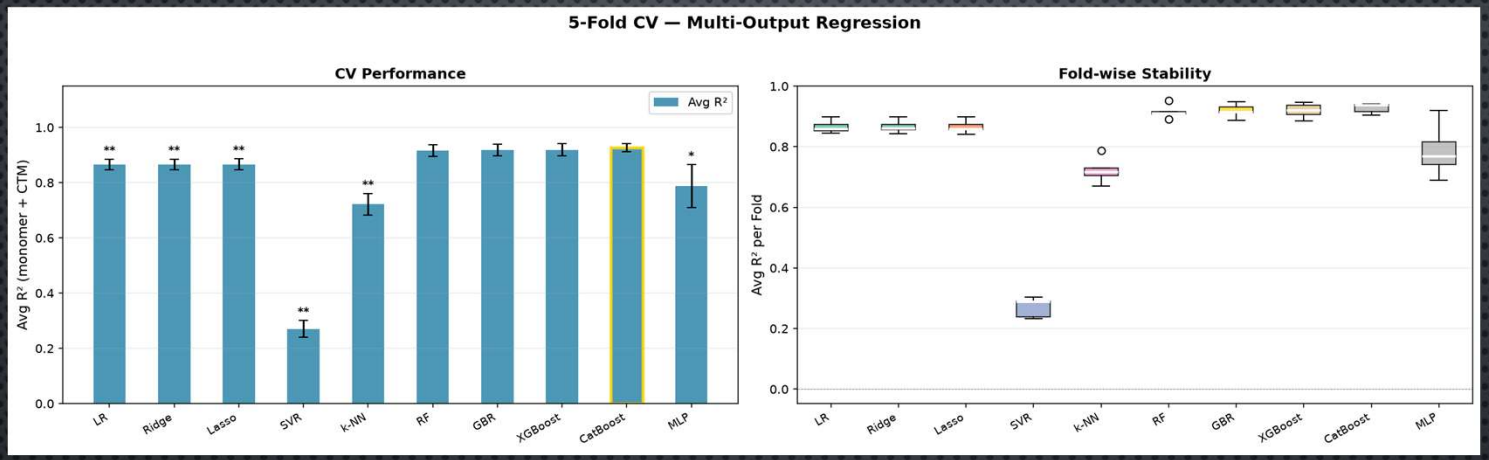
PARITY PLOTS



- Actual versus Predicted values:
  - y_pred = model.predict(X_test)
    - x-axis: actual values (empirical), y_test
    - y-axis: predicted values (theoretical), y_pred

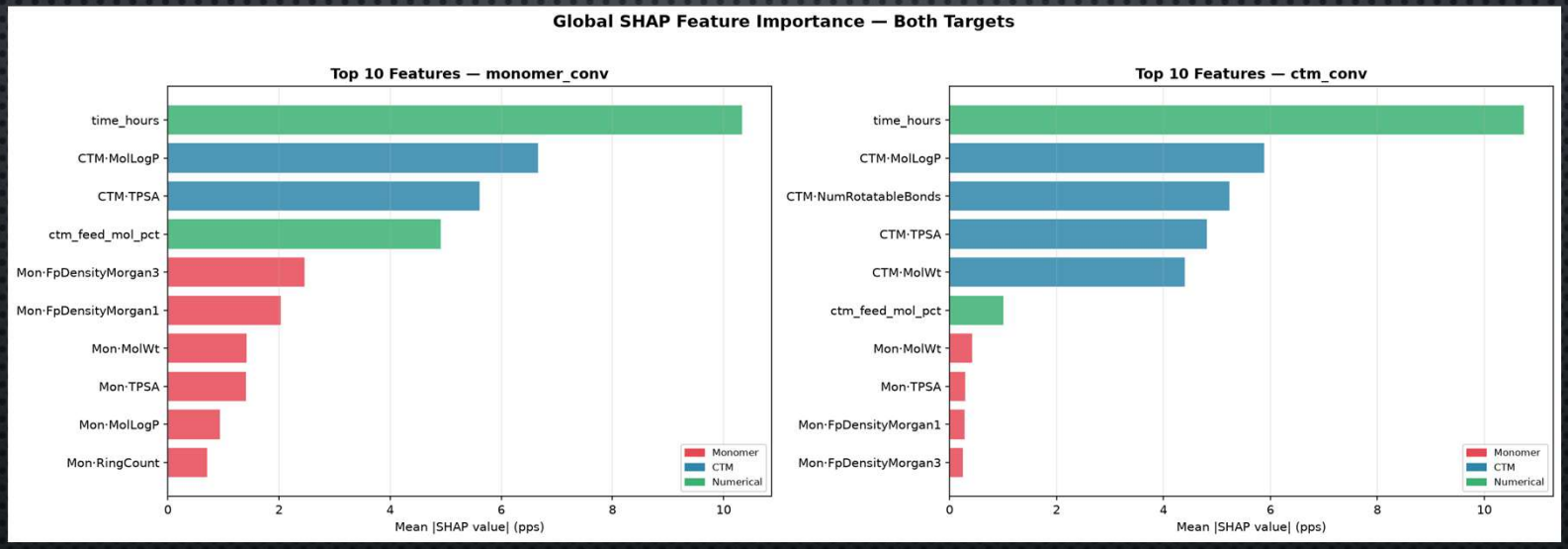
- In layman's terms: how *close* observed values in product yield "match up" with predicted values from the chemical equation and molar quantities of reactants→expected product yield.

CV COMPARISON

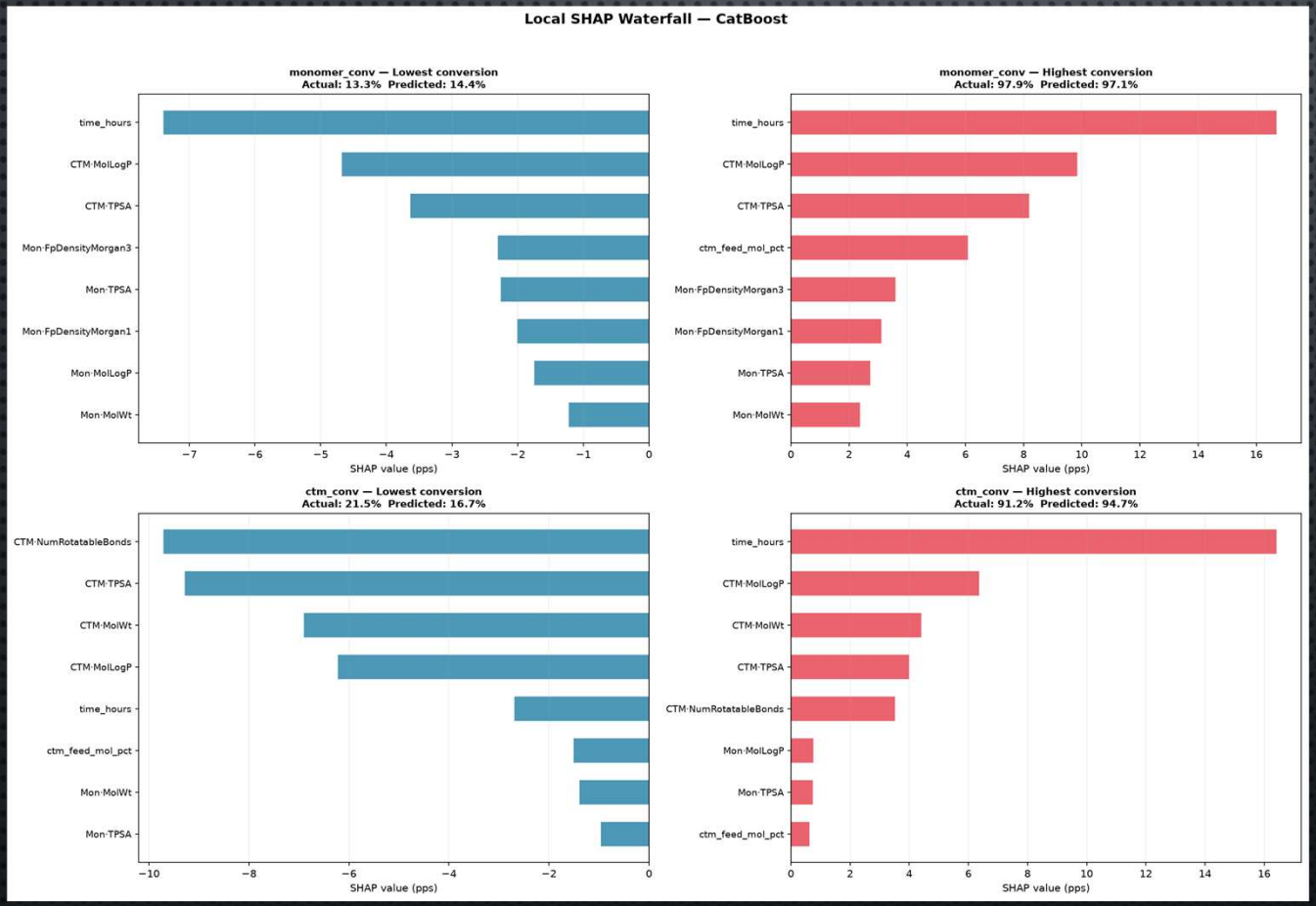


- This is a visualization of a five-fold sampled cross-validation chart.
- When trained and evaluated 5 times as a statistical model, the results are mean-averaged.
- If similar scores are observed across successive empirically observed (trained) <READ: SAMPLED> and are congruent with mutually exclusive <READ: COMPARED AGAINST> (tested) test intervals known as *folds*, then the statistical model is **valid.**

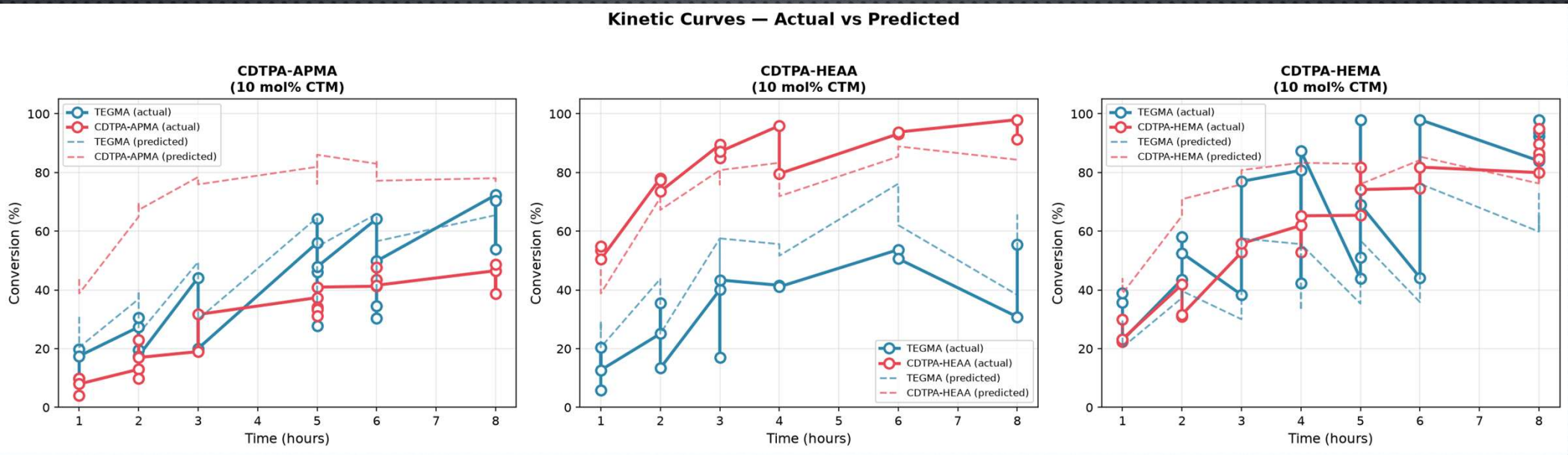
MONOMER NUMBER WAS THE MOST IMPORTANT PREDICTOR OF CONVERSION--GRAPHICAL CONCLUSIONS BELOW:



SHAP WATERFALL GRAPH



KINETIC CURVES GRAPHS



RESULTS SUMMARY CHART OUTPUT DATA

<CTM KINETIC CONVERSION PREDICTION>

Conclusion: Here, the CatBoost software code achieved the best balance of accuracy and precision to validate stability:

- Among target chain transfer monomers <as experimental reactant variables> CDTPA-HEMA, CDTPA-HEAA, CDTPA-APMA that …
- Bind to the lone TEGMA monomer <as an empirically analyzed as a control reactant variable> to form …
- Empirically observed reaction products of molar percentage yield and measurable time per reaction.

model	cv_r2_avg	cv_sd	test_r2_mo	test_r2_ctn	test_r2_avg	rmse_mon	rmse_ctm
XGBoost	0.9188	0.0218	0.9205	0.9578	0.9392	6.354	5.132
CatBoost	0.9272	0.0145	0.9229	0.9538	0.9383	6.259	5.369
GBR	0.9184	0.0206	0.911	0.9587	0.9349	6.724	5.073
RF	0.9158	0.0204	0.9023	0.966	0.9342	7.043	4.608
Ridge	0.8661	0.0189	0.8583	0.9093	0.8838	8.483	7.522
LR	0.8658	0.0188	0.859	0.9086	0.8838	8.465	7.551
Lasso	0.8662	0.0196	0.8577	0.9096	0.8836	8.504	7.509
MLP	0.7869	0.078	0.7865	0.9377	0.8621	10.415	6.234
k-NN	0.7219	0.0385	0.8038	0.8224	0.8131	9.983	10.525
SVR	0.2709	0.0296	0.3285	0.4285	0.3785	18.47	18.881